



Greenhouse Gasses Emissions in World's Economy Report

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Data handling and visualization details

All data manipulations, visualizations, and textual insights were developed within a single R Markdown file. The use of R Markdown enabled integration of R code, Markdown, and LaTeX, ensuring reproducibility, clarity, and transparency throughout the analysis process. The raw data was processed using various R functions, including filtering, grouping, and deriving plotted indicators. The `dplyr` and `tidyr` libraries were employed to clean, transform, and aggregate the data as needed. Visualizations were created using the `ggplot2` and `treemapify` packages. The pdf format was designed for the exercise and is written in LaTeX.

- Chart 1: The year-on-year percentage change in GHG emissions was calculated for the EU27, the Euro area (EA), and global totals. Emissions for the Euro area were derived by summing the totals of the countries within the EA.
- Chart 2: Data on income categories and total populations was sourced from the publicly available World Bank database. After cleaning the data (re-coding country codes and removing incomplete entries) to synchronize it with the EDGAR database, CO₂ emissions per capita were calculated for each income group for the year 2023
- Chart 3: Data mapping countries to continents was collected, focusing solely on individual countries. The percentage of total GHG emissions was calculated for each of the seven continents as well as individual countries. The report's source code can be found on my [GitHub repository](#).

Insights report

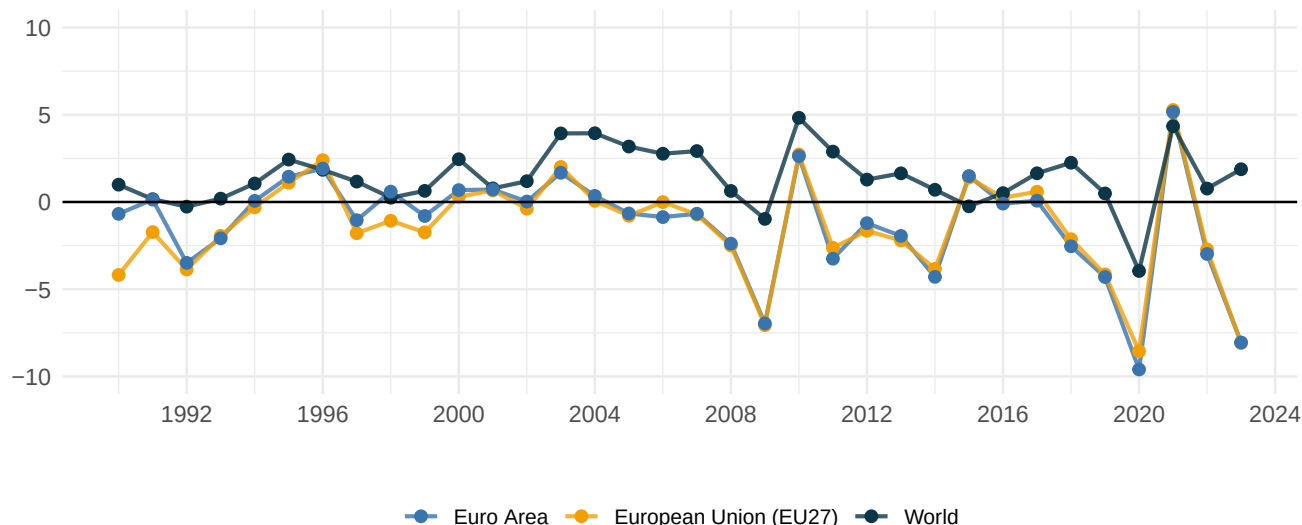
Evolution of GHG growth in the euro area, European Union (EU27) and worldwide

- In the European Union (EU27) and Euro Area, there was a noticeable 8.1% (repetitively 261 Mt CO_{2-eq} and 196 Mt CO_{2-eq}) decrease in total GHG emissions in 2023 compared to 2022. On the other hand, the World GHG emission increased by 1.8% (994 Mt CO_{2-eq}).
- From 1990 to 2023 the EU27 and Euro Area countries showed steady decline in GHG emissions with -1.2% compounded annual growth rate (CAGR) while Total World GHG emissions continue to raise with CAGR equal to 1.5%.



Greenhouse Gasses Emission – Year on year growth rate

Growth rate in %

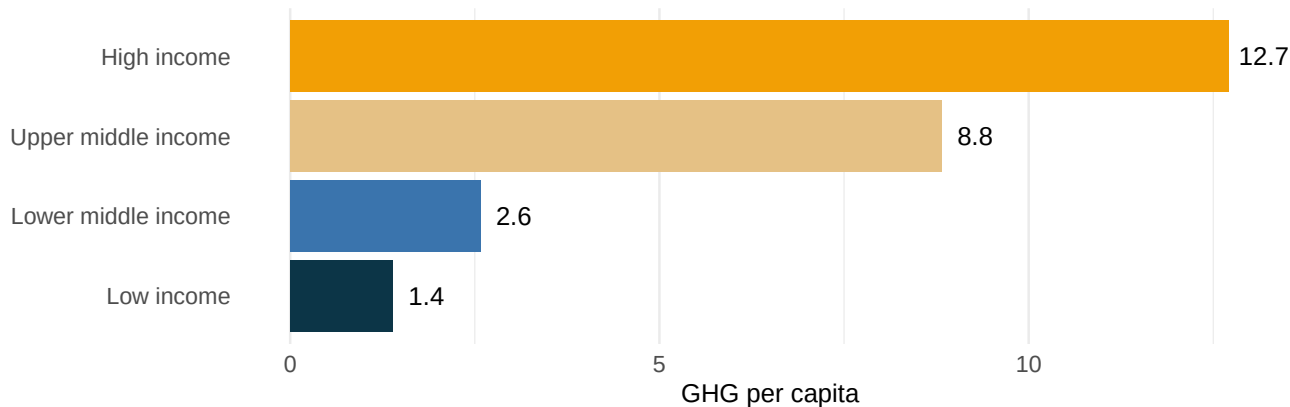


Comparison of countries' GHG emissions per capita aggregated according to the World Bank income groups

- High-income countries emit around 12.4 t of CO_{2-eq} per capita whereas low-income countries emit almost eight times less tonnes of CO_{2-eq} per capita with 1.5 t CO_{2-eq} per capita which highlights disproportional emissions responsibility of wealthier nations.
- There is strong relationship between the country's the Gross Domestic Product (GDP) and the amounts of GHG it produces. Wealthier countries due to their advanced industrialization and economic growth produce more.

Greenhouse Gasses Emission per capita – 2023 income groups comparison

Tonnes of CO_{2-eq} per capita per year



Notes.

- (1) Individual countries were classified based on the latest classification provided by World Bank (<https://datatopics.worldbank.org/world-development-indicators/the-world-by-income-a>).
- (2) The GHG per capita was calculated using population totals from 2023 provided by World Bank (<https://data.worldbank.org/indicator/SP.POP.T>).



Contribution of individual country and continent GHG emissions to total world GHG emissions

- China, the United States, India, Russia and Brazil were the world's largest GHG emitters in 2023. Together they account for 49.8% of global population and 62.7% of global GHG emissions.
- Asia is responsible for the largest share of emissions due to rapid industrialization in countries like China and India. African continent, having 2nd largest share in world population, accounts only for 6% of total of total GHG emissions which highlight.

Contribution of continent and individual country GHG emissions to total World GHG emissions in 2023

Proportion of Total GHG emission in %

